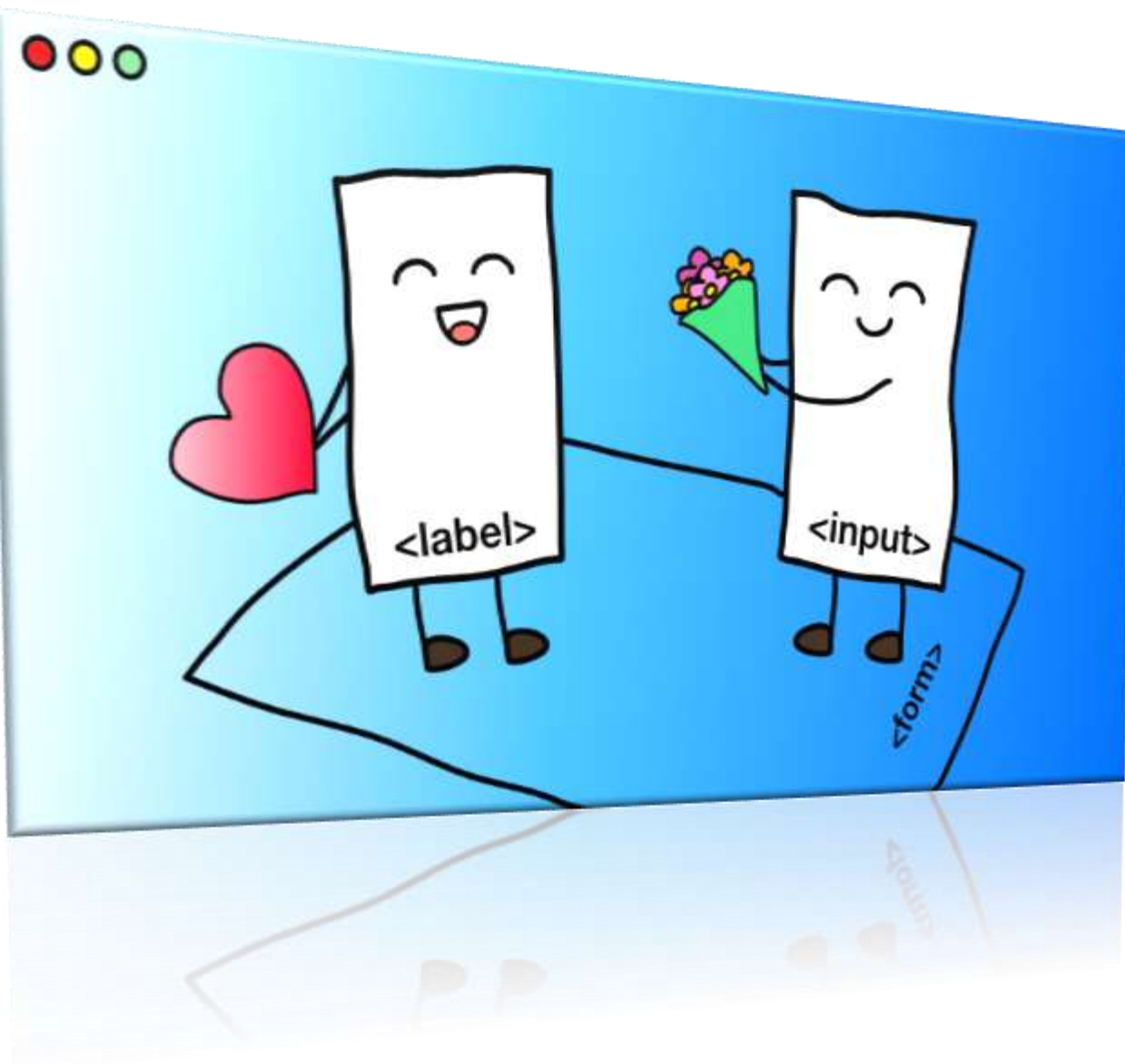




Let's get Started!

Tables and Forms
(Input, Label,
Button)

Today's Topics

❖ Understanding Tables in HTML

❖ Table Structure and Semantic

Elements

❖ Use Cases and Best Practices

for Tables

❖ Role and Structure of Forms

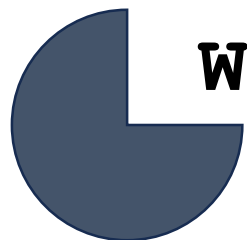
❖ Input Controls and Form

Elements

❖ Labels and Accessibility

❖ Button Types and Functionality





What is a Table in HTML?

- An **HTML table** is a way to represent **structured data** in **row-column** format.
- It is particularly useful for **tabular data** like marksheets, product listings, or schedules.
- The table ensures that data is **readable, organized**, and easy to scan visually.
- Unlike layout tables used in early HTML, modern tables are used strictly for **data**

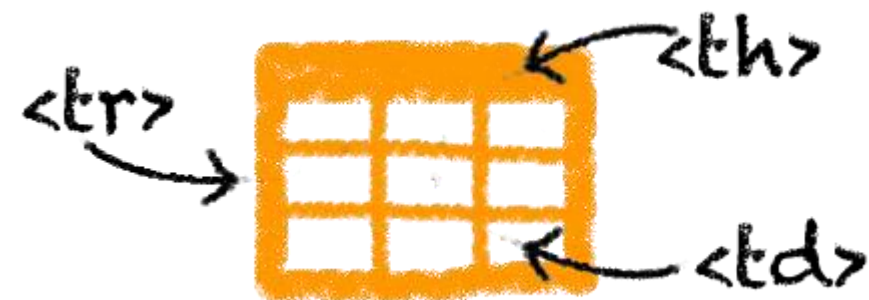


Table Tags Explained

HTML tables are built using a group of elements:

- `<table>`: Defines the table container.
- `<tr>` (Table Row): Represents a row of data.
- `<th>` (Table Header): Defines a header cell, usually bold and centered.
- `<td>` (Table Data): Represents a standard data cell.

Structure Example:

```
<table>
  <tr>
    <th>Subject</th>
    <th>Marks</th>
  </tr>
  <tr>
    <td>Math</td>
    <td>95</td>
  </tr>
</table>
```



Table Attributes and Styling

Legacy Attributes (less preferred today):

- `border`: Adds a visible border.
- `cellspacing`: Space between cells.
- `cellpadding`: Space inside cells.

Modern Styling via CSS:

- `border-collapse: collapse;` for cleaner borders.

- `text-align`, `padding`, `background-color` for readability.

Cell Merging:

- `colspan`: Merge columns.
- `rowspan`: Merge rows.

```
<td
```

```
colspan="2">Merged
```

- ❖ Use CSS for modern responsive tables





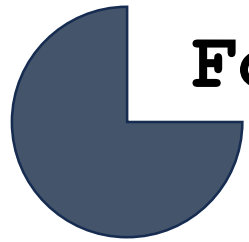
What is an HTML Form?

Structure:

- A **form** is an HTML structure that allows users to **input data** and send it to a **server**.
- Used in applications like:
 - Login/Registration
 - Feedback/Contact forms
 - Surveys, Search boxes
- **action:** Where data is sent
- **method:** How it is sent (GET or POST)

```
<form action="submit.php" method="post">  
  <!-- Form fields here -->  
</form>
```





Form Submission Methods

➤ GET:

- Appends data in the URL
- Less secure
- Suitable for search forms

➤ POST:

- Sends data in the request
- More secure
- Used for login, registration, etc.

❖ Always use POST for sensitive data like passwords.

GET

- Values are contained in the URL;
- Has a length limitation of 255 characters;
- It is often cacheable;
- Supports only string data types;
- Parameters are saved in browser history;
- Results can be bookmarked.

POST

- Values are contained in the message's body;
- Does not have a length limitation ;
- It is hardly cacheable;
- Supports different data types, such as string, numeric, binary, etc;
- Parameters are not saved in browser history;
- Results cannot be bookmarked.

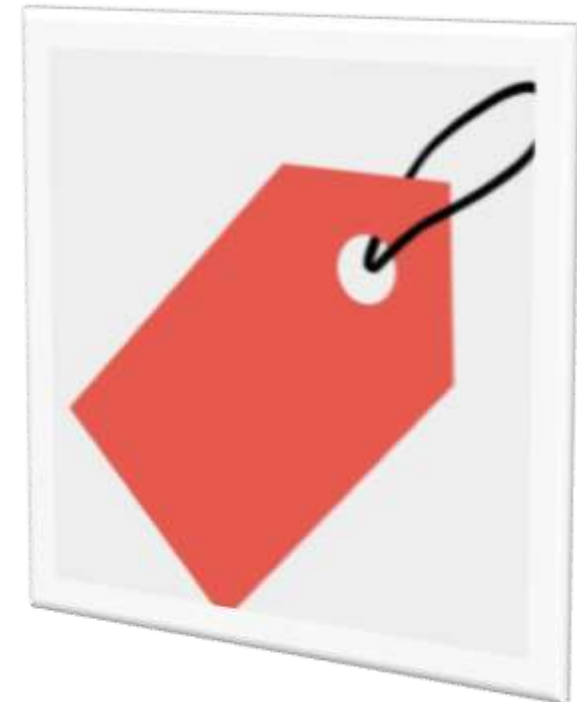


The Input Tag in Depth

- The `<input>` tag is a **versatile form element**.
- Self-closing and supports many types via type attribute:
- text, password, email, number, radio, checkbox, submit

```
<input type="text" name="username">
```

- name: Identifies the field for server-side processing
- id: Links to the `<label>` for accessibility
- placeholder: Guides the user with sample text
- Inputs form the **core** of user interaction in web forms.



Input Types with Meaning

Example:

- **Text:** For short text input

```
<input type="checkbox" name="lang" value="Java"> Java
```

- **Password:** Masks the characters

- **Email:** Validates email pattern

- **Number:** Accepts only numeric values

- **Radio:** Choose one option from a group

- **Checkbox:** Choose multiple options

- **Submit/Reset:** Control form submission

❖ Choosing the right input type improves usability and validation.





The Label Tag – Importance of Accessibility

- The `<label>` tag **describes** a form control.
- Improves usability and screen reader support.
- Linked to input via `for` and `id`.

```
<label for="email">Email:</label>  
<input type="email" id="email" name="email">
```

❖ Labels make forms accessible to users with disabilities.



The Button Tag – Functional Control

❖ HTML buttons can perform different actions.

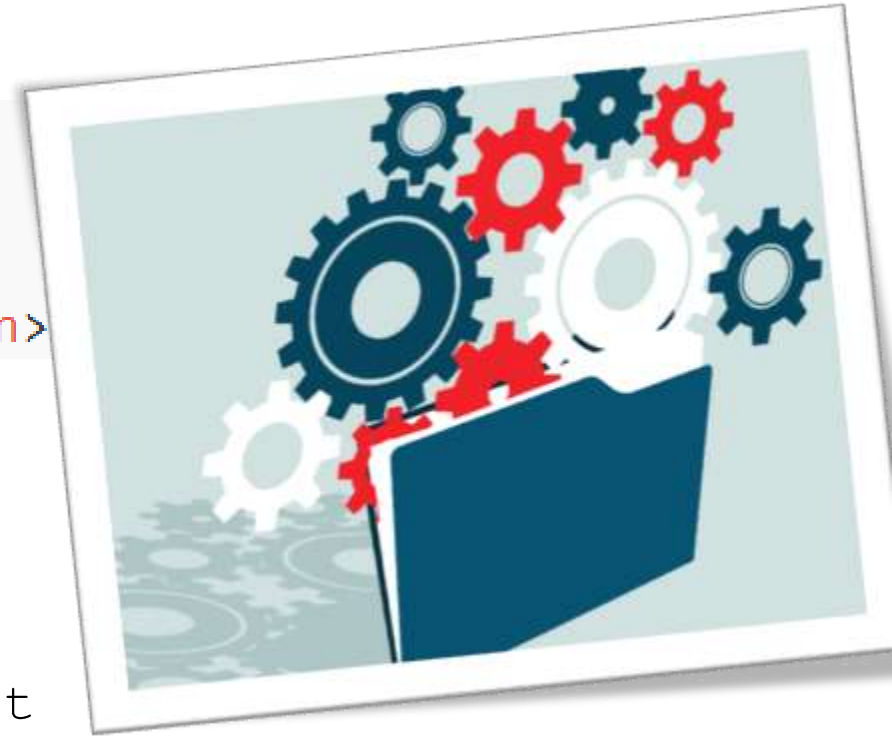
```
<button type="submit">Submit</button>
```

```
<button type="reset">Reset</button>
```

```
<button type="button">Click Me</button>
```

- type="submit": Submits the form
- type="reset": Clears all fields
- type="button": Used with JavaScript

➤ Always define the type attribute to avoid unexpected behavior.



Complete Form Example

```
<form action="login.php" method="post">  
  <label for="user">Username:</label>  
  <input type="text" id="user" name="username"><br><br>  
  
  <label for="pass">Password:</label>  
  <input type="password" id="pass" name="password"><br><br>  
  
  <button type="submit">Login</button>  
</form>
```



Best Practices for Tables and Forms

- Use **semantic tags** (<th> for headers) in tables
- Avoid using tables for layout design – use CSS
- Always pair <input> with <label>.
- Use placeholders and default values to guide users
- Keep forms short and simple – reduce cognitive load.
- Validate input on **both client and server sides**.



That's All for Today
Guys!!

Any Queries?

